Revision Module

Temporal Consistency, Historical Regimes, and the Completed Sovereign Cross-Section

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Abstract

This revision extends the empirical design of the preliminary 30-sovereign cross-section by distinguishing numerical sample completion from temporal and institutional comparability. Three previously unavailable observations have been recovered: the Netherlands' ten-year government bond yield, the United States' federal-funds target range, and Egypt's overnight deposit policy rate. Their inclusion changes the nominal availability counts from 30 CDS observations, 29 bond yields, and 28 policy or banking reference rates to a nominally complete 30-observation inventory for each variable. This arithmetic completion does not establish that the observations are synchronous, that their instruments are institutionally equivalent, or that the underlying economic relationships remained stable over the observation window.

The revision therefore introduces an observation-time vector, a measure of within-country temporal dispersion, and an institutional-regime description. It adds historical case notes for Egypt, the Netherlands, and the United States, and distinguishes the legacy 27-case estimates from estimates that must be recomputed on the completed sample. The recovered United States target range is retained as a range rather than silently converted into an unqualified scalar. The Egyptian overnight deposit rate is identified separately from the overnight lending, main-operation, and discount rates. The Netherlands yield is reported with a source-discrepancy warning. The resulting dataset is numerically more complete, but its interpretation requires explicit temporal and institutional controls.

The revision ends with "The End"

1 Revision Scope and Central Distinction

The original empirical study examined the cross-sectional relationship between five-year sovereign credit default swap (CDS) spreads, ten-year government bond yields, and central-bank or banking-system reference rates for thirty sovereigns. The initial inventory contained thirty CDS observations, twenty-nine ten-year yields, twenty-eight policy or reference rates, and twenty-seven complete cases.

The three previously missing fields have now been recovered. The arithmetic of sample completion is:

$$N_{\text{CDS}} = 30, \tag{1}$$

$$N_{\text{yield}} = 29 + 1 = 30, \tag{2}$$

$$N_{\text{rate}} = 28 + 2 = 30. \tag{3}$$

Consequently, the nominal complete-case count becomes

$$N_{\text{complete}} = 30, \tag{4}$$

subject to verification of the remaining country-level records and the instrument definitions.

This result must not be confused with preservation of the original economic state. In particular,

$$30 = 30 - 3 + 3 \tag{5}$$

describes restoration of the number of observations, whereas

$$0 = 3 - 3 \tag{6}$$

describes the elimination of the three missing fields. Neither identity establishes temporal synchronisation, institutional equivalence, or structural stability.

The central methodological distinction of this revision is therefore:

> A complete cross-section is not necessarily a contemporaneous cross-section, and a contemporaneous cross-section is not necessarily a structurally homogeneous cross-section.

2 Recovered Observations and Measurement Conventions

2.1 Netherlands: ten-year government bond yield

The Netherlands ten-year government bond yield is recorded as

$$Y_{\mathrm{NL},10Y} = 3.34\%. \tag{7}$$

The observation date is 10 November 2026. A historical-data table reports 3.34% for that date. Other market-data providers report values close to 3.33% for the same date. The World Government Bonds historical page available during this revision displayed a last update of 4 November 2026, and therefore did not independently verify the 10 November observation.

The value 3.34% is consequently retained as the recovered working observation, with a source-discrepancy flag. Before final estimation, the research archive should preserve the provider, retrieval timestamp, quoted benchmark, closing-time convention, and unrounded value. The value must not be represented as an official Dutch Treasury fixing unless such a fixing has been independently established.

2.2 United States: federal-funds target range

The Federal Reserve's target range for the federal-funds rate is recorded as

$$r_{\mathrm{US}}^{\mathrm{target}} \in [3.75\%, 4.00\%]. \tag{8}$$

The Federal Reserve's policy-rate page identifies this range as the latest target range on 29 October 2026. The effective federal-funds rate is a different observed market rate: the Federal Reserve's H.15 release reports 3.88% for 2–6 November 2026.

The target range and effective rate must not be conflated. If the empirical specification requires one scalar target-rate proxy, the midpoint may be computed transparently:

$$r_{\mathrm{US}}^{\mathrm{mid}} = \frac{3.75 + 4.00}{2} = 3.875\%. \tag{9}$$

This midpoint is a derived statistic, not a separately announced Federal Reserve instrument. The preferred data architecture retains the lower bound, upper bound, midpoint, effective rate, and their respective dates as separate fields. The main regression must state which field is used.

2.3 Egypt: overnight deposit rate

The Central Bank of Egypt's Monetary Policy Committee announced on 29 October 2026 a 100-basis-point increase in its key rates. The resulting rates were:

$$r_{\mathrm{EG}}^{\mathrm{deposit}} = 20.00\%, \tag{10}$$

$$r_{\mathrm{EG}}^{\mathrm{lending}} = 21.00\%, \tag{11}$$

$$r_{\mathrm{EG}}^{\mathrm{main}} = 20.50\%, \tag{12}$$

$$r_{\mathrm{EG}}^{\mathrm{discount}} = 20.50\%. \tag{13}$$

For a single policy-rate field, this revision uses the overnight deposit rate, 20.00%, as the explicitly identified working convention. This selection must be applied consistently or accompanied by sensitivity estimates using the other CBE rates. The four rates are distinct instruments and must not be averaged or substituted without an explicit rule.

Table 1: Recovered observations and instrument definitions

Sovereign	Variable	Working value	Observation / decision date	Qualification
Netherlands	Ten-year government bond yield	3.34%	10 Nov. 2026	Provider discrepancy; benchmark and close convention to be archived
United States	Federal-funds target range	3.75–4.00%	29 Oct. 2026	Retain range; midpoint 3.875% is derived
United States	Effective federal-funds rate	3.88%	2–6 Nov. 2026	Separate market rate, not the target range
Egypt	Overnight deposit rate	20.00%	29 Oct. 2026	Working scalar policy-rate convention
Egypt	Overnight lending rate	21.00%	29 Oct. 2026	Distinct CBE instrument
Egypt	Main-operation rate	20.50%	29 Oct. 2026	Distinct CBE instrument
Egypt	Discount rate	20.50%	29 Oct. 2026	Distinct CBE instrument

3 Temporal Structure of the Cross-Section

3.1 Observation time is variable-specific

The original design treated the sovereign as the unit of observation. The revised design retains that unit but recognises that the three principal variables may have different observation dates. Let

$$\mathbf{t}_i = \left(t_i^{CDS}, t_i^{Y}, t_i^{r}\right) \tag{14}$$

denote the observation-time vector for sovereign i, where the components refer respectively to the CDS spread, ten-year yield, and policy-rate variable.

The within-sovereign temporal dispersion is defined as

$$\Delta_i = \max\left\{t_i^{CDS}, t_i^{Y}, t_i^{r}\right\} - \min\left\{t_i^{CDS}, t_i^{Y}, t_i^{r}\right\}. \tag{15}$$

Here, dates are represented in calendar time. Where timestamps are available, the preferred implementation uses timestamp differences and records time zones and market closing conventions.

A small Δ_i indicates closer temporal alignment, but it does not prove that the observations were measured under identical market conditions. A large Δ_i warns that the variables may describe different information sets or policy environments.

3.2 Decision time, effective time, and observation time

The data architecture must distinguish at least three dates:

1. **Announcement or decision date**: when the relevant authority announced a policy decision.
2. **Effective date**: when the decision became operational.
3. **Market observation date**: when the market yield, CDS spread, or effective overnight rate was measured.

For the United States, the target-range decision date and the effective date should be stored separately from the daily effective federal-funds rate. For Egypt, the MPC announcement date should be distinguished from the date on which the new rates became effective, if the source records those dates separately. A policy decision is not itself a market observation.

3.3 Institutional and historical time

Let $g_i(t)$ denote the institutional regime applicable to sovereign i at time t. It may encode, for example, monetary-union membership, exchange-rate arrangements, the legal status of the policy instrument, and the institutional allocation of monetary authority.

Let $h_i(t)$ denote a broader historical-state descriptor. This may include major monetary reforms, financial crises, sovereign restructurings, war-related fiscal changes, sanctions, or other structural events.

The observed cross-section is therefore better represented as

$$\mathcal{D} = \left\{ S_i(t_i^{CDS}), Y_i(t_i^{Y}), r_i(t_i^{r}), \mathbf{t}_i, g_i(\cdot), h_i(\cdot) \right\}_{i=1}^{30}. \tag{16}$$

This representation does not claim that every historical factor has been quantified. It makes explicit that the economic meaning of an observation depends partly on its institutional and historical context.

4 Historical Case Notes: Three Monetary Trajectories

4.1 Egypt: monetary sovereignty and changing monetary standards

The Central Bank of Egypt's historical account records the development of the Egyptian pound within a changing monetary architecture. A decree in 1834 provided for a currency system based on gold and silver, and the Egyptian pound was minted and offered for circulation in 1836. Monetary reform in 1885 established the gold standard as the basis of the Egyptian monetary system. On 2 August 1914, convertibility into gold was suspended and Egyptian banknotes became the legal monetary instrument.

These transitions demonstrate that the same currency denomination can operate under materially different monetary standards. A policy rate observed under a contemporary fiat-money framework cannot be interpreted as though it were institutionally equivalent to an interest rate observed under a metallic standard or a fixed external monetary arrangement.

For the present cross-section, the relevant implication is not that the historical regimes should be collapsed into a single numerical control. Rather, the current policy-rate observation must be interpreted in light of the contemporary monetary framework, while longer-run historical analysis should encode regime transitions explicitly.

4.2 Netherlands: the guilder, Amsterdam finance, and monetary union

The Bank of Amsterdam was founded in 1609 by the municipal authorities. Its public giro and payments functions formed part of the institutional infrastructure supporting commerce and financial settlement. The guilder subsequently occupied an important place in Dutch monetary history.

The Netherlands adopted the euro on 1 January 1999. Euro banknotes and coins entered circulation on 1 January 2002, and the dual-circulation period ended on 28 January 2002. The Netherlands is therefore a useful case for distinguishing national sovereign bond markets from national monetary-policy autonomy. Dutch government bonds remain national sovereign liabilities, but monetary policy is conducted within the Eurosystem.

Accordingly, the Netherlands' ten-year sovereign yield must not be paired with a fictitious independent Dutch national policy rate. The relevant common monetary-policy instrument is an appropriate ECB policy rate, with the exact facility identified. This also applies to other euro-area members in the sample.

4.3 United States: central-bank architecture and the fiscal-monetary relationship

The Federal Reserve System was established by the Federal Reserve Act of 1913. Its institutional history includes the Great Depression, wartime interest-rate controls, and the Treasury–Federal Reserve Accord of March 1951. The Accord marked an important institutional separation between government debt management and monetary policy.

This history illustrates why a nominal policy rate cannot be interpreted independently of the fiscal-monetary relationship. The same numerical interest rate can arise under different institutional constraints and different central-bank reaction functions.

For the present dataset, the federal-funds target range is the announced policy instrument. The effective federal-funds rate is a market outcome within the operating framework. They are related, but they are not identical variables.

5 Revised Sample Availability and Data Architecture

Table 2: Revised nominal data availability as of 10 November 2026

Variable	Available observations
Five-year sovereign CDS spread	30
Ten-year government bond yield	30
Policy or banking reference rate	30
Nominal complete cases	30

The revised counts are nominal availability counts. They do not imply that all thirty policy-rate fields are already harmonised as identical instruments. In particular, the dataset contains different types of monetary variables: central-bank target rates, target ranges, deposit facility rates, and potentially banking-system reference rates. Each observation must retain its instrument metadata.

The recommended data schema is:

Field	Definition
sovereign	Standardised country identifier
cds_5y_bp	Five-year sovereign CDS spread, basis points
cds_date	CDS observation date and, where available, timestamp
yield_10y_pct	Ten-year sovereign government bond yield, percent per annum
yield_date	Bond-yield observation date and timestamp
yield_benchmark	Bond identifier, benchmark convention, currency, and source
policy_value_pct	Selected policy-rate scalar, if the instrument is scalar
policy_lower_pct	Lower bound of a policy target range, where applicable
policy_upper_pct	Upper bound of a policy target range, where applicable
policy_midpoint_pct	Derived midpoint, never confused with the announced range
policy_effective_pct	Effective market rate, stored separately from the target
policy_instrument	Exact instrument name and operating definition
policy_decision_date	Announcement or decision date
policy_effective_date	Date on which the decision became effective
policy_observation_date	Date of the recorded rate observation
source_url	Direct source record or archived source reference
retrieval_timestamp	Date and time the researcher retrieved the source
regime_code	Institutional monetary-regime classification
source_quality_flag	Verification, rounding, or provider-discrepancy flag

6 Implications for Descriptive Statistics and Regression

The original paper reported its correlation matrix and preliminary regressions using twenty-seven complete cases. Those results remain valid only as descriptions of that original estimation sample, subject to the original data and calculation conventions. They must not be relabelled as thirty-country estimates.

In particular, the previously reported correlation between ten-year yields and the policy or banking reference-rate variable was

$$\rho_{Y,r}^{(27)} = 0.989. \tag{17}$$

The corresponding variance inflation factor was approximately

$$\mathrm{VIF}^{(27)} = \frac{1}{1-(0.989)^2} \approx 45.6. \tag{18}$$

These are legacy 27-case diagnostics. They must be recomputed after incorporating the recovered observations and resolving instrument definitions. The same rule applies to the reported logarithmic-CDS correlations and the original Model A and Model B estimates.

The revised empirical workflow should report, at minimum:

1. the legacy 27-case estimates, labelled as the baseline;
2. the 30-case estimates after source verification and instrument coding;
3. a sensitivity estimate excluding observations with unresolved source or instrument comparability flags;
4. a sensitivity estimate using the US target-range midpoint rather than the target range as a categorical interval;
5. an Egypt sensitivity estimate using the overnight lending rate or main-operation rate instead of the overnight deposit rate;
6. a specification that distinguishes euro-area sovereigns from sovereigns with independent national monetary-policy instruments.

For each specification, the exact estimation sample must be stated. The completed sample should not be used to conceal changes in the underlying sample composition. If a model uses a single scalar policy variable, the construction of that scalar must be stated before the regression is estimated.

6.1 Temporal sensitivity

Where sufficient daily observations are available, the preferred extension is to construct a temporally aligned panel or matched-date cross-section. For each sovereign, the researcher may select a common reference date or a pre-specified maximum matching window. The matching rule must be chosen before inspecting the regression results.

A simple matched-date rule can be represented as

$$\left|t_i^k - t^\star\right| \leq \tau_k, \qquad k \in \{CDS, Y, r\}, \tag{19}$$

where $t^\star$ is the common reference date and τ_k is the permitted variable-specific tolerance. Observations outside the tolerance are flagged or excluded from the matched-date specification rather than silently carried forward.

A cross-section assembled from asynchronous observations can remain useful descriptively, but it should be called an asynchronous reference cross-section. A panel design is required to investigate the dynamics of transmission, temporal precedence, or regime-dependent adjustment.

7 Revised Limitations and Identification

The completed inventory removes three missing fields, but it does not remove the following limitations:

1. **Asynchronous measurement.** CDS spreads, bond yields, and policy variables may refer to different dates and market closing times.
2. **Instrument heterogeneity.** Policy target ranges, deposit facility rates, lending rates, main-operation rates, and banking reference rates are not automatically interchangeable.
3. **Institutional heterogeneity.** Euro-area sovereigns share a monetary authority while issuing separate national sovereign debt.
4. **Historical regime changes.** Monetary institutions and fiscal-monetary relationships have changed materially over time.

5. **Source disagreement.** Market-data providers may report slightly different benchmark yields or closing values.

6. **Endogeneity.** Policy rates and sovereign yields may respond to the same shocks that affect CDS spreads.

7. **Small-sample sensitivity.** Thirty observations remain a small cross-section, especially when institutional indicators or interaction terms are introduced.

The cross-sectional regressions remain associational. Neither sample completion nor the inclusion of historical narratives supplies a causal identification strategy. Causal claims require a separate design based on credible exogenous policy variation, instruments, natural experiments, or other defensible identifying restrictions.

8 Revised Conclusion

The empirical inventory now contains thirty nominal observations for each of the three principal variables. The Netherlands ten-year yield, the United States federal-funds target range, and Egypt's overnight deposit rate have been recovered and documented. This is a meaningful improvement in data completeness.

The principal methodological conclusion, however, is that numerical completion is not equivalent to historical or institutional homogeneity. The Netherlands' sovereign yield exists within a monetary union; the United States target range must be distinguished from the effective federal-funds rate; and Egypt's several policy rates represent different instruments within its monetary framework. Historical experience further demonstrates that the institutional meaning of monetary variables changes over time.

The original 27-case estimates must therefore remain identified as baseline estimates until the final 30-case dataset has been source-checked and all descriptive statistics and regressions have been recomputed. The completed dataset is a stronger starting point, not a licence to claim that all observations describe one perfectly synchronous economic state.

The governing methodological principle of this revision is:

> Sample completeness is a property of the data inventory. Temporal comparability is a property of the observation process. Institutional comparability is a property of the measurement definitions. Structural stability is an empirical hypothesis. None follows automatically from the others.

References for the Revision

References

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